Object Detection & Tracking of Tennis Players and Balls

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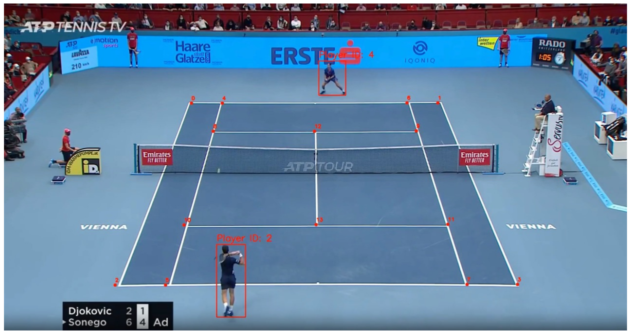
Overview of the System

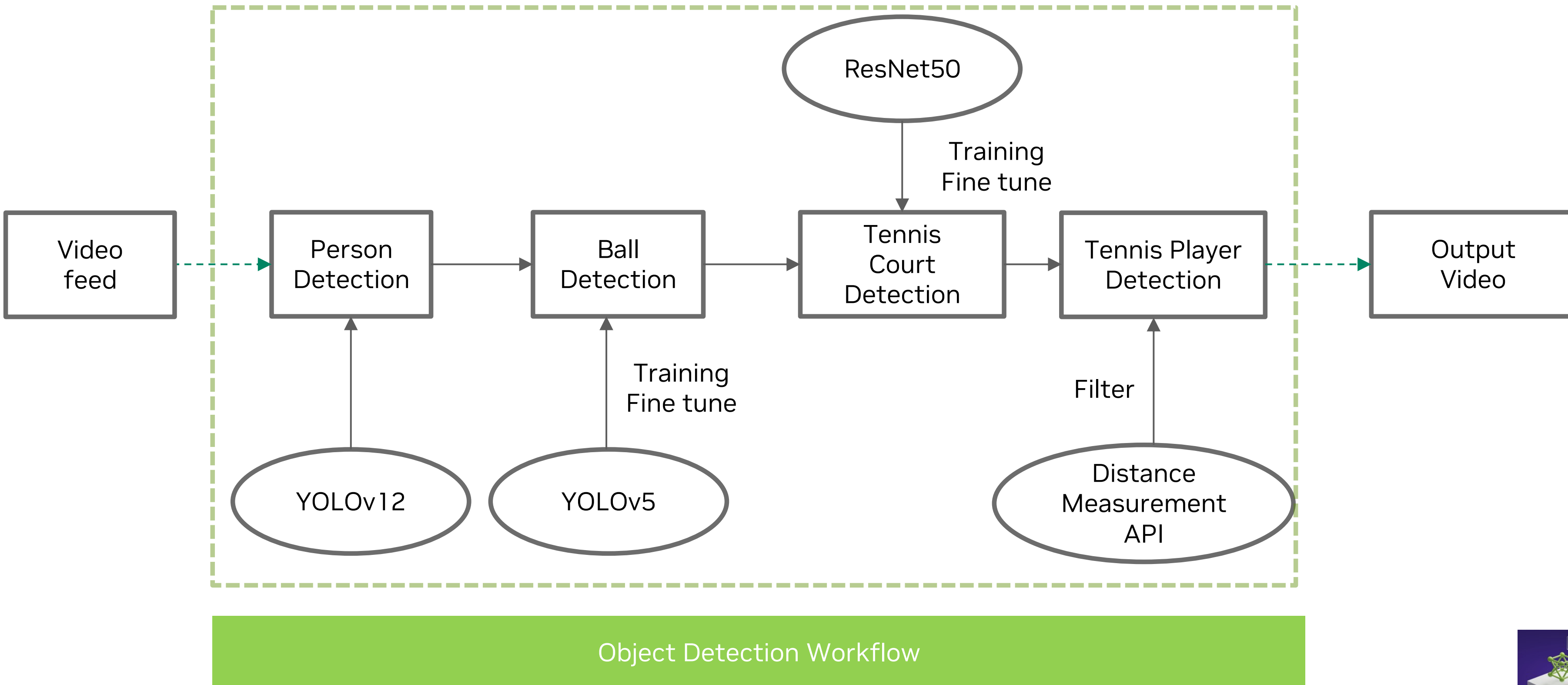
Tennis Player and Balls Detection

- Key Components:
    - Object Detection: Using YOLOv12 and YOLOv5 for player and ball detection.
    - Court Detection: Using TennisCourtDetector to identify 14 keypoints of the tennis court.
    - Tracking: Real-time tracking of players and the ball across video frames.
    - Post-Processing: Classical computer vision methods for refining keypoint predictions.



Solution Architecture

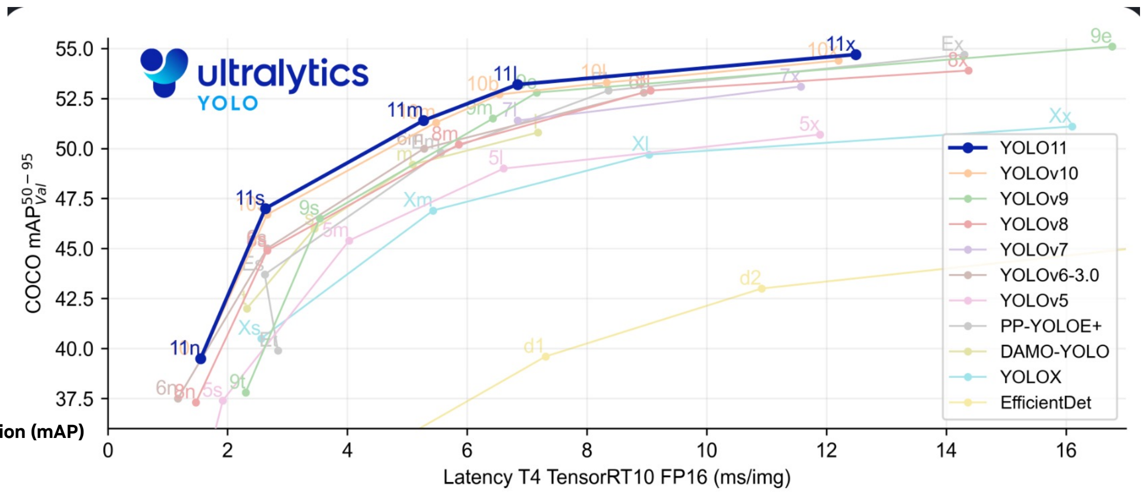
Tennis Player and Balls Detection



Implementation

YOLOv12 and YOLOv5 for Object Detection

- YOLOv12:
  - Advanced architecture for high-speed.
  - Suitable for general object detection and robust under dynamic environments.
- YOLOv5:
  - Best for detecting small objects like tennis balls due to multi-scale features.
  - Chosen for ball detection due to its robustness and high accuracy.



Mean Average Precision (mAP)

link

Implementation

Training YOLOV5 and Results

- Dataset:
  - Training set: 428 images
  - Validation set: 100 images
  - Test set: 50 images
- Fine-tune YOLOV5 for 100 epochs
- Results:
  - mAP50: 0.645 (mean average precision at 50% IoU threshold)
  - mAP50-95: 0.266 (mean average precision at multiple IoU thresholds between 50% and 95%)
  - Precision: 0.794 (predicted positive instances were actually correct)
  - Recall: 0.525 (how many of the true positive instances were correctly predicted)
- Speed:
  - Preprocess: 0.2ms per image
  - Inference: 10.4ms per image
  - Postprocess: 2.7ms per image

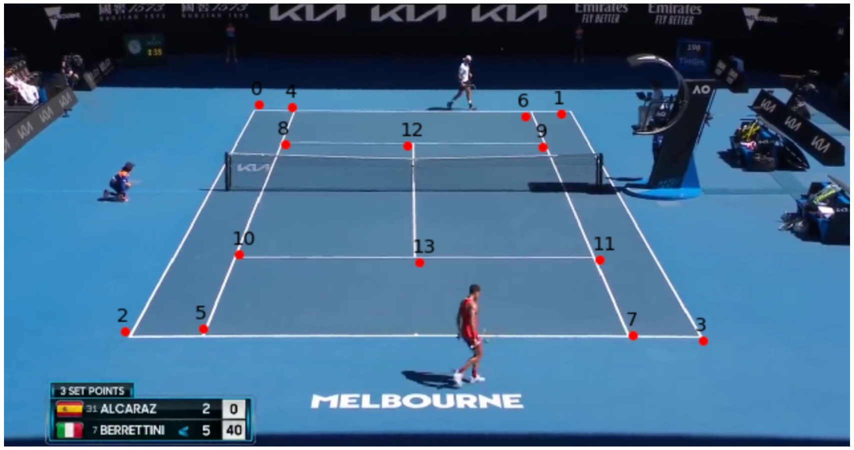
```
100 epochs completed in 1.138 hours.
Optimizer stripped from runs/detect/train/weights/last.pt, 172.6MB
Optimizer stripped from runs/detect/train/weights/best.pt, 172.6MB

Validating runs/detect/train/weights/best.pt...
Ultralytics 8.3.51 🚀 Python-3.10.12 torch-2.5.1+cu121 CUDA:0 (Tesla T4, 15102MiB)
YOLOv5l6u summary (fused): 393 layers, 85,972,308 parameters, 0 gradients, 137.0 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 4/4 [00:01<00:00,  2.06it/s]
                   all        100        101      0.794      0.525      0.645      0.266
Speed: 0.2ms preprocess, 10.4ms inference, 0.0ms loss, 2.7ms postprocess per image
Results saved to ESC[1mruns/detect/trainESC[0m
💡 Learn more at https://docs.ultralytics.com/modes/train
```

Implementation

TennisCourtDetector and Court Keypoint Detection

- TennisCourtDetector:
  - Deep learning model to detect 14 keypoints of the tennis court.
  - Keypoints include baselines, service lines, net intersections, etc.
  - Reference: link
- Post-Processing:
  - Keypoint refinement using classical methods (line detection).
  - Homography-based correction for geometric accuracy.



Implementation

Training ResNet50 Model and Results

- Tennis Court Key Points
  - 14 key points for detecting tennis court boundaries
- Dataset:
  - 8841 images [link]
  - 75% used for training 25% used for validation
  - Image resolution: 1280 x 720, resized to 244 x 244 pixels
  - Normalization
- ResNet Backbone:
  - Used for feature extraction in image classification
- Fine-tuning:
  - Fine-tune ResNet-50 for 50 epochs
  - As iterations increase, the loss decreases
  - Save the model in .pth format for further use

```
Epoch 0, iter 0, loss: 14244.68359375
Epoch 0, iter 10, loss: 14543.7041015625
Epoch 0, iter 20, loss: 13650.517578125
Epoch 0, iter 30, loss: 14030.8642578125
Epoch 0, iter 40, loss: 13343.619140625
Epoch 0, iter 50, loss: 13222.3369140625
Epoch 0, iter 60, loss: 12664.3349609375
Epoch 0, iter 70, loss: 12113.822265625
Epoch 0, iter 80, loss: 12411.8623046875
Epoch 0, iter 90, loss: 11664.3017578125
Epoch 0, iter 100, loss: 11559.98046875
Epoch 0, iter 110, loss: 10293.25
Epoch 0, iter 120, loss: 9871.798828125
Epoch 0, iter 130, loss: 10517.828125
Epoch 0, iter 140, loss: 10093.8125
Epoch 0, iter 150, loss: 9563.6435546875
Epoch 0, iter 160, loss: 9305.9951171875
Epoch 0, iter 170, loss: 8040.826171875
Epoch 0, iter 180, loss: 7964.0068359375
Epoch 0, iter 190, loss: 7656.208984375
Epoch 0, iter 200, loss: 7792.2490234375
Epoch 0, iter 210, loss: 7728.93603515625
Epoch 0, iter 220, loss: 7098.0146484375
Epoch 0, iter 230, loss: 7172.091796875
Epoch 0, iter 240, loss: 6553.89404296875
...
Epoch 49, iter 790, loss: 0.45051679015159607
Epoch 49, iter 800, loss: 0.7321450710296631
Epoch 49, iter 810, loss: 0.34632608294487
Epoch 49, iter 820, loss: 1.4448059797286987
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

```
torch.save(model.state_dict(), "keypoints_model_50.pth")
```

References

Tennis Player and Balls Detection

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